The BrainCapture Transformer Model for EEG

William Lehn-Schiøler, Magnus Guldberg Pedersen, and Benjamin Kock Fazal

P27 - EEGPT: A General-Purpose Transformer for EEG Analysis



Motivation

Automation tools like AI can assist in EEG diagnostics, but the scarcity of publicly labelled EEG data limits their development. Recent advances in self-supervised learning and transformer models, like GPT-4, show that unlabeled data can capture inherent structures and context. We propose a **general-purpose EEG transformer**, pre-trained on the large, unlabeled TUH EEG corpus (TUEG), which can be fine-tuned with smaller labelled datasets for tasks such as seizure detection, abnormality detection, and artifact removal.

Methods

By using **Self-Supervised Learning (SSL),** the BrainCapture Transformer model for EEG can

- Learn general structures and underlying patterns from massive amounts of unlabelled data
- Recognize important features, such as brain activity changes associated with seizure or interictal findings
- Become more accurate while requiring fewer manually annotated EEG recordings

Results

Dataset	Task	# Classes	F1-Score
BrainCapture Bhutan	Artifacts	5	0.983 ± 0.002
Siena Scalp EEG	Seizures	2	0.902 ± 0.008
TUH EEG Seizure	Seizures	2	0.972 ± 0.002
TUH EEG Abnormal	Abnormalities	2	0.949 ± 0.014
TUH EEG Slowing	Abnormalities	3	0.884 ± 0.045



References

The following references are relevant to this work:

1. SPEED: **Scalable preprocessing of EEG data** for self-supervised learning, Gjølbye et al., 2024.
2. BENDR: using **transformers and a contrastive self-supervised learning** task to learn from massive amounts of EEG data, Kostas et al., 2021.
3. wav2vec 2.0: A framework for **self-supervised learning of speech representations**, Baevski et al, 2020.
4. **Attention** is all you need, Vaswani et al, 2017.

3rd International Conference on Artificial Intelligence in Epilepsy and Neurological Disorders

March 3 - 6, 2025
Breckenridge, CO, USA